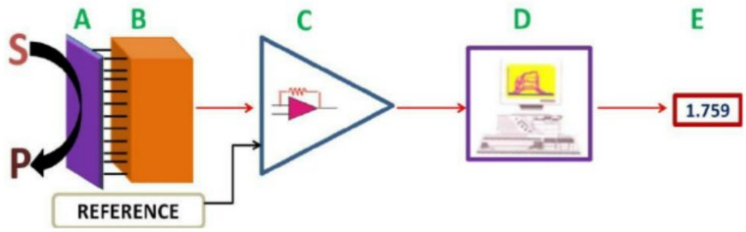
EE6301 Quiz-1 Date: 25/09/2024

Each question worth 5 points, you have 50 minutes. Please choose the best answer.

1. Given below is the diagram of the structure of a biosensor. What is the function of B?



1. 下图是生物传感器的结构示意图。B 的功能是什么?
A. 将生物信号转换为电信号
B. 过滤背景噪声
C. 放大低强度信号
D. 作为生物界面进行主动感测
E. 提高检测信号的读出质量

A. To convert biological signals into electrical signals
B. To filter out background
C. To amplify low-intensity signals
D. To serve as a bio-interface for active sensing
E. To improve the readout quality of the detection signal

2. 以下关于荧光的说法中，哪一项是错误的?
A. 具有较长荧光寿命的样品将提供更高的光谱分辨率
B. 二向色镜的截止波长应位于吸收波长和发射波长之间
C. 荧光显微镜必须包含可以激发样品的光源
D. 双光子荧光显微镜利用能量较低的两个光子形成更高分辨率的图像

2. Which of the following regarding fluorescence is wrong?
A. A sample with a longer fluorescence lifetime will provide a higher spectral resolution
B. The cutoff wavelength of dichroic mirrors should be between the absorption and emission wavelengths.
C. Fluorescence microscopy must include a light source that can excite the sample.
D. Two-photon fluorescence microscopes utilize two photons with lower energy to form a higher resolution image

3. 光学谐振器在许多生物传感器中起到了重要作用，以下哪一项不能作为生物传感器中的光学谐振器?

3. Optical resonators have played an important role in many biosensors, which of the following cannot serve as an optical resonator in biosensor?
A. Two-mirrors cavity
B. Surface plasmon resonance
C. Optical fibers
D. Raman spectroscopy

A. 双镜腔（Two-mirrors cavity）
B. 表面等离子共振（Surface plasmon resonance）
C. 光纤（Optical fibers）
D. 拉曼光谱（Raman spectroscopy） （正确答案，拉曼光谱通常不用于光学谐振）

4. 以下关于生物传感器的哪项陈述是不正确的?

A. 特异性（Specificity）意1味着它可以在众多目标中区分出特定目标

4. Which of the following statements is not true for biosensors?
A. Specificity means that it can distinguish one specific target among the rest
B. Sensitivity is used to determine how low of analyte contention can detected
C. Linearity of the response should be higher the better
D. Evanescent field is used for detection when light penetrates from higher to lower refractive index

B. 灵敏度（Sensitivity）用于确定分析物浓度的最低检测限
C. 响应的线性度越高越好
D. 消逝场（Evanescent field）用于当光从高折射率介质进入低折射率介质时的检测 （错误，应该是从低折射率介质进入高折射率介质） （正确答案）

5. The working principle of an optical blood oxygen sensor based on?
A. Total internal reflection when light hits red blood cell in fingertip
B. Absorption and reflection difference between oxygen/less oxygen red blood cells
C. Fluorescence emission between oxygen/less oxygen red blood cells
D. The ratio between the red LED and green LED emission intensity

5. 光学血氧传感器的工作原理是什么?
A. 当光照射到指尖的红细胞时发生全内反射
B. 氧合/缺氧红细胞的光吸收和反射差异 （正确答案）
C. 氧合/缺氧红细胞之间的荧光发射
D. 红色 LED 和绿色 LED 发射强度的比值

6. 以下关于微阵列和微流控生物传感器的说法，哪一项是不正确的？
A. 光刻（Photolithography）和喷墨打印（Inkjet printing）是常见的制造工具。
B. **芯片实验室（Lab-on-a-chip）**意味着将微阵列和微流控系统集成到一个生物传感器中，使其具有多个功能。
C. 微阵列生物芯片的目的是同时检测多种分析物。

6. Which is not correct for microarray and microfluidic biosensors?
   A. Photolithography and inkjet printing are common tools for fabrication.
   B. Lab-on-a-chip means the integration of microarray and microfluid system with multiple functions in one biosensor.
   C. The purpose of microarray biochips is to detect multiple analytes at the same time. D. 生物芯片可用于测量细胞、DNA、蛋白质。
   D. Biochips can be used to measure cells, DNA, proteins
   E. Biochips are usually made of paper in nowadays due to its strong stretchability.
   E. 现代生物芯片通常由纸制成，因为它具有很强的可拉伸性。 （错误，生物芯片通常由硅、玻璃或聚合物材料制成，而不是纸） （正确答案）

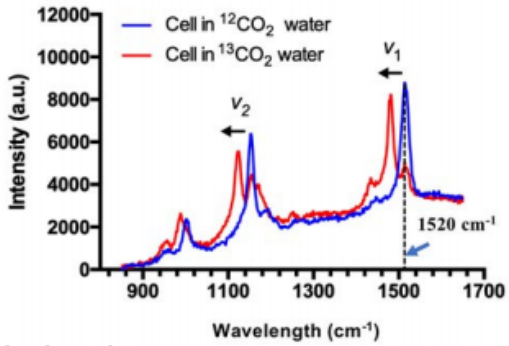
7. Which is not related to the working principle for label-free optical biosensors?
   A. Changes in terms of fluorescence intensity
   B. Changes in terms of reflectivity
   C. Changes in terms of refraction index
   D. Changes in terms of wavelength shift
   E. Changes in terms of resonance frequency

7. 以下哪一项与无标记光学生物传感器（label-free optical biosensors）的工作原理无关？
A. 荧光强度的变化（ 正确答案，与无标记传感器无关）
B. 反射率的变化
C. 折射率的变化
D. 波长偏移的变化
E. 共振频率的变化

8. 下图显示了一种光合细胞传感器在含 CO_2 和不含 CO_2 的水中培养时的拉曼光谱。拉曼位移从 1520 cm^{-1} 变化到 1510 cm^{-1}。计算这个位移，以 "nm"（纳米）为单位。

8. The figure shows the Raman spectra of a photosynthetic cell sensor that was cultured in water with CO2 and without CO2. The changes of Raman shift is from 1520cm^{-1} to 1510 cm^{-1}. Calculate this shift in the unit of "nm".
   a) $1x10^6$ nm
   b) $1x10^7$ nm
   c) $2x10^{-6}$ nm
   d) $1x10^{-8}$ nm
   e) $2x10^6$ nm

$$\Delta\lambda = \frac{\Delta\tilde{\nu} \times 10^7}{\tilde{\nu}_1 \times \tilde{\nu}_2}$$



9. 以下哪个参数取决于 B 模式超声成像中的空间脉冲长度（spatial pulse length）？

9. Which of the following parameters is dependent on spatial pulse length in B-mode ultrasound imaging?
   A. Penetration depth
   B. Transverse resolution
   C. Imaging speed
   D. Axial resolution

A. 穿透深度（Penetration depth）
B. 横向分辨率（Transverse resolution）
C. 成像速度（Imaging speed）
D. 轴向分辨率（Axial resolution） （正确答案，轴向分辨率受空间脉冲长度影响）

10. 以下哪一项不影响超声成像中的穿透深度（penetration depth）？

10. Which of the following does NOT affect the penetration depth in ultrasound imaging?
   A. Frequency of acoustic wave
   B. Spatial pulse lenght
   C. Attenuation coefficient
   D. Input ultrasound intensity

A. 声波频率（Frequency of acoustic wave）
B. 空间脉冲长度（Spatial pulse length）
C. 衰减系数（Attenuation coefficient） （正确答案，影响超声波信号的强度但不会直接决定穿透深度）
D. 输入超声波强度（Input ultrasound intensity）

11. Which one of the following information is NOT needed to calculate transverse resolution of a light microscope?
A. Magnification
B. Wavelength of light
C. Acceptance angle of objective lens
D. Immersion medium

11. 以下哪项信息在计算光学显微镜的横向分辨率时是不需要的?
A. 放大倍率（Magnification）
B. 光的波长（Wavelength of light）
C. 物镜的接受角（Acceptance angle of objective lens）
D. 浸没介质（Immersion medium）　（正确答案，不影响分辨率的计算）

12. Which is the advantage of ultrasound imaging compared with optical coherence tomography
A. Larger penetration depth
B. Higher resolution
C. Cross-sectional imaging mode
D. Ability to measure distance

12. 与光学相干断层扫描（OCT）相比，超声成像的优势是什么?
A. 更大的穿透深度（Larger penetration depth）　（正确答案）
B. 更高的分辨率（Higher resolution）
C. 横截面成像模式（Cross-sectional imaging mode）
D. 具有测量距离的能力（Ability to measure distance）

13. For Abdomen, brain, and ocular imaging, what are the best modality, respectively?
A. OCT, MRI, Ultrasound
B. MRI, OCT, Ultrasound
C. MRI, ultrasound, OCT
D. Ultrasound, MRI, OCT

13. 在腹部、大脑和眼部成像中，哪种成像方式最佳? （依次对应）
A. OCT（光学相干断层扫描）、MRI（磁共振成像）、超声（Ultrasound）
B. MRI、OCT、超声
C. MRI、超声、OCT　（正确答案，MRI 适用于大脑，超声适用于腹部，OCT 适用于眼部）
D. 超声、MRI、OCT

14. What is the main signal source of MRI in human body
A. Hydrogen
B. Carbon
C. Oxygen
D. Nitrogen

14. MRI（磁共振成像）在人体中的主要信号来源是什么?
A. 氢（Hydrogen）　（正确答案）
B. 碳（Carbon）
C. 氧（Oxygen）
D. 氮（Nitrogen）

15. Which is the advantage of ultrasound image over MRI?
A. whole organ imaging
B. Higher resolution
C. Lower cost
D. Cross-sectional imaging mode

15. 与 MRI 相比，超声成像的优势是什么?
A. 整个器官成像（Whole organ imaging）
B. 更高的分辨率（Higher resolution）
C. 更低的成本（Lower cost）　（正确答案，超声设备比 MRI 设备便宜且更易获取）
D. 横截面成像模式（Cross-sectional imaging mode）

16. 以下关于拉曼散射（Raman scattering）的说法，哪一项是不正确的?
16. Which sentence about Raman is not correct?
A. Photons are mostly absorbed in Raman process to generate scattering
B. Raman shift of 5000 cm^{-1} is within the infrared region.
C. Raman sensors require the molecule to be polarizable
D. Raman scattering is the scattering interaction between light and particles

A. 在拉曼过程中，光子大部分被吸收以产生散射　（错误，拉曼散射是非弹性散射，光子不会被吸收）
B. 5000 cm^{-1} 的拉曼位移位于红外区域内。（正确）
C. 拉曼传感器要求分子可极化（正确）
D. 拉曼散射是光与粒子之间的散射相互作用（正确）

17. Which of the following describing analysis tools in biosensors and systems is wrong?
A. Bio-images provide a better understanding of two-dimensional information than spectrum.
B. Smart phone can be used to capture two-dimensional information from biosensors.
C. Electronic nose is a device that measures chemicals in surrounding environment.
D. ART test for covid-19 utilizes localized surface plasmon resonance
E. Distributed fiber sensor is designed to measure several types of different signals

17. 以下关于生物传感器和系统分析工具的描述，哪一项是错误的?
A. 生物影像（Bio-images）比光谱数据能更好地提供二维信息理解（正确）
B. 智能手机可用于从生物传感器中捕捉二维信息　（错误，智能手机的成像能力有限，无法直接用作高精度生物传感器）　（正确答案）
C. 电子鼻（Electronic nose）是一种可以测量环境中化学物质的设备（正确）
D. COVID-19 的 ART 测试利用了局部表面等离子共振（正确）
E. 分布式光纤传感器（Distributed fiber sensor）可用于测量不同类型的信号（正确）

18-20. 一种光谱测量系统用于检测癌症相关蛋白，该系统使用 2 cm 厚的比色皿进行测量。蛋白质在 532 nm 处有吸收峰，并在 620 nm 处表现出荧光发射。光信号由光谱仪收集，该光谱仪的光栅分辨率为 500。2

18.-20. An optical spectrometry system is used to detect cancer related protein in a curette of 2 cm thick. The protein expresses an absorption peak at 532 nm and fluorescence emission at 620nm. The optical signals were then collected by a spectrometer with a grating of resolving power of 500.

18. 该蛋白的浓度是多少?

18. What is the concentration of the protein? Assume OD=2, molecular absorption coefficient=2.0 x 10^{-2}mol^{-1}Lcm^{-1}.
    A. 0.025 mol.L^{-1}
    B. 0.05 mol.L^{-1}
    C. 0.25 mol.L^{-1}
    D. 5.0 mol.L^{-1}
    E. 2.5 mol.L^{-1}

光密度（OD）= 2

摩尔吸收系数（Molecular Absorption Coefficient）= 2.0×10^{-2} **mol⁻¹ L cm⁻¹**

比色皿厚度（l）= 2 cm

比尔-朗伯定律（Beer-Lambert Law）：$A = \varepsilon C l$

最终公式： $A = \varepsilon c L$

- A: 吸光度 (Absorbance)
- c: 浓度 (Concentration)
- L: 光程长度 (Path length)

$$C = \frac{A}{\varepsilon l} = \frac{2}{(2.0 \times 10^{-2}) \times 2}$$

$$C = 5.0 \text{ mol.L}^{-1}$$

19. After measuring the emission intensity for various protein concentration, a linear relationship between the protein concentration and intensity was obtained. The slope of the calibration curve is 3x10^{10} unit/M, and the standard deviation is 1255.25 unit. What is the limit of detection (M)?
    A. 0.125 μM
    B. 0.150 μM
    C. 0.250 μM
    D. 1.250 μM

19. 计算该蛋白的检测限（LOD）。

校准曲线斜率 = 3×10^{10} **unit/M**

标准偏差（Standard Deviation）= 1255.25 unit

LOD 计算公式： $LOD = \frac{3 \times \text{Standard Deviation}}{\text{Slope}}$

$$LOD = \frac{3 \times 1255.25}{3 \times 10^{10}}$$

$$= 0.125 \ \mu\text{M}$$

20. Calculate the smallest difference in wavelengths (nm) at peak 620 nm that can be resolved by using in the first-order diffraction pattern.
    A. 1.24 nm
    B. 1.48 nm
    C. 2.48 nm
    D. 0.62 nm

20. 计算在 620 nm 处使用一级衍射图案（first-order diffraction pattern）可以解析的最小波长差（单位：nm）。

光栅的分辨率（resolving power）**R = 500**

波长 λ = 620 nm

公式：

$$\frac{\lambda}{\Delta\lambda} = R$$

计算：

$$\Delta\lambda = \frac{\lambda}{R} = \frac{620}{500}$$

$$= 1.24 \text{ nm}$$

Name: ***Matriculation No:***

1.	**2.**	**3.**	**4.**	**5.**
A	*A*	*D*	*B*	*B*
6.	**7.**	**8.**	**9.**	**10.**
E	*A*	*A*	*D*	*B*
11	**12**	**13**	**14**	**15**
A	*A*	*D*	*A*	*C*
16	**17**	**18**	**19**	**20**
A	*E*	*X*	*A*	*A*